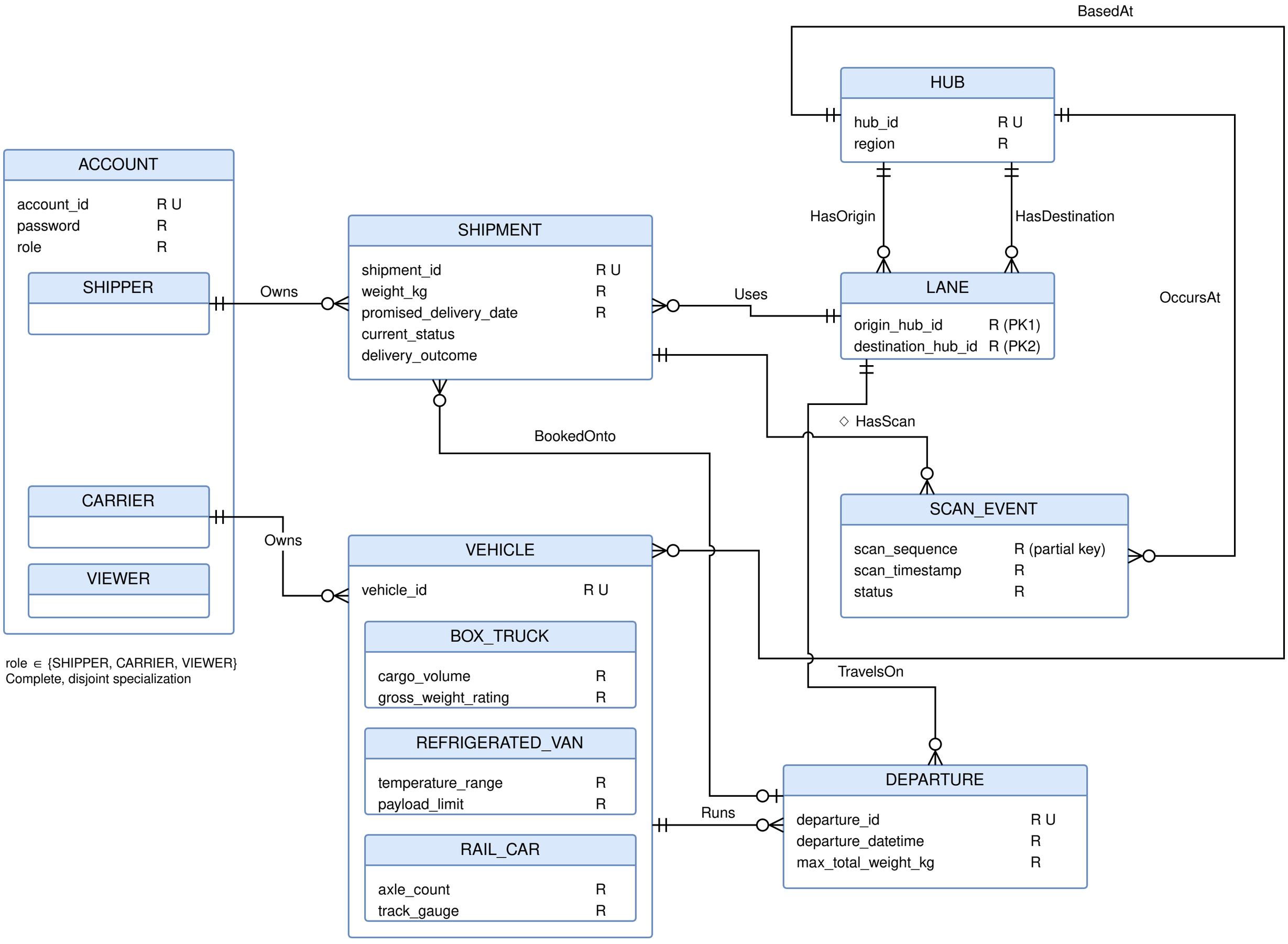

role ∈ {SHIPPER, CARRIER, VIEWER}
Complete, disjoint specialization

Vehicle specialization: disjoint; completeness not specified.

BUSINESS RULES

- HUB.region ∈ {Northeast, Southeast, Midwest, Southwest, West}.
- The LANE key is the ordered pair (origin_hub_id, destination_hub_id). The hubs must differ.
- BookedOnto assumes at most one departure per shipment. Each departure may carry many shipments.
- The booking procedure checks the matching lane and remaining weight capacity atomically.
- The diamond beside HasScan marks the identifying relationship. Each scan requires one shipment.
- The SCAN_EVENT key is (shipment_id, scan_sequence). The sequence is unique within its shipment.
- Scan insertion triggers enforce chronological timestamps and non-regressing status.
- Scan insertion maintains current_status and records delivery_outcome as ON_TIME or LATE on delivery.